

DMPC for Multi-Drone Motion Planning with Dynamic Goals

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Problem & Solution

Gap: Current DMPC assumes static goals

Innovation: Extend to moving targets in [gym-pybullet-drones](#)

Technical Approach

- [Bézier curve](#) trajectories
- [On-demand collision avoidance](#) (50% faster than BVC)
- [Event-triggered replanning](#)
- [Distributed QP](#) at 5 Hz

Extension: Adapt optimization for dynamic goal tracking

Expected Results

Base paper (Luis et al. RAL 2020):

- 90%+ success, 30 agents
- 50% travel time reduction
- 20 Hz control with 20 drones

Our goal: Match performance with moving targets

Implementation

1. Base DMPC in simulator
2. Moving goal cost function
3. Performance benchmarking

Reference: Luis et al., "Online Trajectory Generation with Distributed Model Predictive Control for Multi-Robot Motion Planning" *IEEE RA-L* 2020